CSCI 3060U Phase 3: Front End Testing

Games Distribution System

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November 2026

GitHub: https://github.com/dhruvht612/CSCI-3060U_Project

We are not willing to share our Front End with other students, so we understand we won't be eligible for the bonus marks.

Running the program

The Front End is written in C++17. To compile it, run this from the top folder of the repo in a Bash terminal (we used Git Bash on Windows, but Linux/macOS work too):

```
mkdir -p build
g++ -std=c++17 -Wall -Wextra -Iinclude src/*.cpp -o build/games-distribution
```

The program takes the three input files and then the name of the daily transaction file to write, in this order:

```
build/games-distribution data/currentaccounts.txt data/availablegames.txt \
    data/gamescollection.txt transout.atf
```

You can type commands in directly, or feed in a file of commands with <:

```
build/games-distribution data/currentaccounts.txt data/availablegames.txt \
    data/gamescollection.txt outputs/TC01.atf < inputs/TC01.txt
```

When the commands come from a file, the program prints each line it reads after the prompt. That way the saved output looks the same as what you'd see typing it in.

The commands are login, logout, create, delete, sell, buy, refund, addcredit, list (admin only) and search.

Running the tests

From the top folder of the repo:

```
bash scripts/run_tests.sh
bash scripts/validate_tests.sh
```

`run_tests.sh` compiles the program and runs it once for every file in `inputs/`. For each test it saves the terminal output to `outputs/TCxx.out` and the daily transaction file to `outputs/TCxx.atf`.

`validate_tests.sh` uses `diff` to compare each `.atf` file with `expected/TCxx.etf` and each `.out` file with `expected/TCxx.out`. It prints which tests passed or failed (and the differences for any that failed), then adds a one-line summary to `outputs/test_history.log`.

We also have 21 extra tests for things the main tests don't check directly, like account balances after a buy or refund and what happens with a broken input file. They need Python 3:

```
python3 scripts/supplemental_tests.py --label some-name
```

How the tests are set up

All 54 tests use the same three data files in `data/`. Each test has:

- `inputs/TCxx.txt`: the commands typed in, one per line
- `expected/TCxx.out`: what should show up on the screen
- `expected/TCxx.etf`: the daily transaction file it should write

`tests/TEST_CATALOG.md` has a short description of every test. TC01-TC46 are from Phase 1 and TC47-TC54 test the new SEARCH transaction.

Changes from the Phase 1 feedback

We changed our tests to address each of the TA's comments on Phase 1:

Feedback	What we changed
The output should show what the user will see, not "T0 …, Expected Output: …"	Our expected output files were descriptions of what should happen. We replaced them with the actual screen output (prompts, what's typed, and the responses), and the scripts compare them with `diff`. The descriptions moved to `tests/TEST_CATALOG.md`.
Daily transaction files are not input files	`inputs/` now only has the command files. Transaction files only show up as expected results (`.etf`) and actual results (`.atf`).
Couldn't tell what most of the files in `inputs/` were for	Every test used to have four input files (commands, users, games, collections), so `inputs/` had 216 files. Now all tests share one set of data files and there's just one input file per test.
Random files with just 00 inside	A lot of our expected transaction files only had `00` in them. We changed the 00 line to the handout's format (`00`, username, account type, credit), e.g. `00 admin AA 005000.00`, and fixed the program to match (runs 7 and 8 in the failure log).

We worked out the credit in each new 00 line by hand: we started from the user's balance in `currentaccounts.txt`, then applied any buys, refunds and added credit from that test.

The SEARCH update

SEARCH was added as a new requirement. It lists all games and users whose names contain the search text, but non-admin users shouldn't see admins who aren't selling any games.

How we implemented it:

- Any logged-in user can search. It asks for the search text, and blank text is rejected.
- Matching isn't case-sensitive.
- Games show the name, seller and price. Users show the name and account type.
- If you're not an admin, an admin account only shows up if it has a game for sale.
- Games put up for sale in the same session don't show up until the next session, the same as with buying.
- Search doesn't write anything to the daily transaction file.

TC47-TC54 test this. We wrote them before writing the code, so they failed in run 5 and passed in run 6.

Failure log

We kept every failure we saw, even after fixing it. TC01-TC54 are the main tests (listed in `tests/TEST_CATALOG.md`). S01-S21 are extra tests in `scripts/supplemental_tests.py`. H01 was a problem with one of our scripts.

Test runs

Run	What changed before this run	Main tests	Extra tests
1	First run on our Phase 2 code (most transactions were still stubs)	30/46	7/21
2	Implemented the transactions (see fixes below)	46/46	21/21
3	Reran everything	46/46	21/21
4	Ran from a clean copy of the repo	46/46	-
5	Added TC47-TC54 for SEARCH, before writing any SEARCH code	46/54	-
6	Implemented SEARCH	54/54	21/21
7	Redid the tests from the Phase 1 feedback (no code changes yet)	0/54	5/21
8	Fixed the 00 record and input echo	54/54	21/21

Runs 1-6 used our first test runner (written in Python). Their outputs are saved in `archive/phase3_iterations1-6_evidence/`. Runs 7-8 used `run_tests.sh` and `validate_tests.sh`; the full output is in `outputs/history/` and the summaries are in `outputs/test_history.log`.

Failures

Test	What it was testing	What was wrong with the output	Error in the code / test	What we changed	Result after fix
TC08	Valid Create User	Run 1: the create was rejected, so the .etf only had 00 (the 01 record missing)	createUser() was still a stub that always returned false	Wrote createUser(): checks the name and type, rejects duplicates, adds the account with 0 credit and writes the 01 record	Pass (runs 2 and 3)
TC12	Username Exactly 15 Characters	Run 1: the create was rejected, so the .etf only had 00 (the 01 record missing)	createUser() was still a stub that always returned false	Wrote createUser(): checks the name and type, rejects duplicates, adds the account with 0 credit and writes the 01 record	Pass (runs 2 and 3)
TC13	Valid Delete User	Run 1: the delete was rejected, so the .etf only had 00 (the 02 record missing)	deleteUser() and removeUser() were empty stubs	Wrote deleteUser(): removes the account and the user's games for sale, writes the 02 record	Pass (runs 2 and 3)
TC17	Valid Sell	Run 1: the sell was rejected, so the .etf only had 00 (the 03 record missing)	sell() was still a stub that always returned false	Wrote sell(): checks permission, price and name, writes the 03 record	Pass (runs 2 and 3)
TC19	Maximum Game Price	Run 1: the sell was rejected, so the .etf only had 00 (the 03 record missing)	sell() was still a stub that always returned false	Wrote sell(): checks permission, price and name, writes the 03 record	Pass (runs 2 and 3)
TC21	Game Name Exactly 25 Characters	Run 1: the sell was rejected, so the .etf only had 00 (the 03 record missing)	sell() was still a stub that always returned false	Wrote sell(): checks permission, price and name, writes the 03 record	Pass (runs 2 and 3)
TC25	Transaction on Newly Listed Game	Run 1: the sell was rejected, so the .etf only had 00 (the 03 record missing)	sell() was still a stub that always returned false	Wrote sell(): checks permission, price and name, writes the 03 record	Pass (runs 2 and 3)
TC26	Valid Buy	Run 1: the buy was rejected, so the .etf only had 00 (the 04 record missing)	buy() was a stub, and the games/collections files were never actually read	Wrote the file loaders and buy(): checks seller, ownership and credit, moves the money, writes the 04 record	Pass (runs 2 and 3)
TC30	Exact Credit Purchase	Run 1: the buy was rejected, so the .etf only had 00 (the 04 record missing)	buy() was a stub, and the games/collections files were never actually read	Wrote the file loaders and buy(): checks seller, ownership and credit, moves the money, writes the 04 record	Pass (runs 2 and 3)
TC32	Valid Refund	Run 1: the refund was rejected, so the .etf only had 00 (the 05 record missing)	refund() and adjustCredit() were stubs	Wrote refund(): checks both accounts, moves the credit, writes the 05 record	Pass (runs 2 and 3)
TC36	Valid Add Credit - Standard User	Run 1: the addcredit was rejected, so the .etf only had 00 (the 06 record missing)	addCredit() and adjustCredit() were stubs, and nothing tracked the $1000-per-session limit	Wrote addCredit() with a per-account session total, writes the 06 record	Pass (runs 2 and 3)
TC37	Valid Add Credit - Admin	Run 1: the addcredit was rejected, so the .etf only had 00 (the 06 record missing)	addCredit() and adjustCredit() were stubs, and nothing tracked the $1000-per-session limit	Wrote addCredit() with a per-account session total, writes the 06 record	Pass (runs 2 and 3)
TC39	Add Exactly $1000	Run 1: the addcredit was rejected, so the .etf only had 00 (the 06 record missing)	addCredit() and adjustCredit() were stubs, and nothing tracked the $1000-per-session limit	Wrote addCredit() with a per-account session total, writes the 06 record	Pass (runs 2 and 3)
TC41	Multiple Credit Additions to $1000	Run 1: the addcredit was rejected, so the .etf only had 00 (both 06 records missing)	addCredit() and adjustCredit() were stubs, and nothing tracked the $1000-per-session limit	Wrote addCredit() with a per-account session total, writes the 06 record	Pass (runs 2 and 3)
TC42	Multiple Credit Additions Over $1000	Run 1: the addcredit was rejected, so the .etf only had 00 (the 06 record missing)	addCredit() and adjustCredit() were stubs, and nothing tracked the $1000-per-session limit	Wrote addCredit() with a per-account session total, writes the 06 record	Pass (runs 2 and 3)
TC46	Daily Transaction File Formatting	Run 1: the create was rejected, so the .etf only had 00 (the 01 record missing)	createUser() was still a stub that always returned false	Wrote createUser(): checks the name and type, rejects duplicates, adds the account with 0 credit and writes the 01 record	Pass (runs 2 and 3)
S01	Very large credit amount	Run 1: the program crashed with "Error: stoll" instead of rejecting the amount	std::stoll throws on a number that big and nothing caught it	Rewrote the money parser in Money.h to reject amounts that are too large	Pass (runs 2 and 3)
S03	Buying the same game twice; balances after a buy	Run 1: second buy wasn't rejected properly and balances were wrong (admin should be 4985.00, seller1 515.00)	buy() was a stub, and the games/collections files were never actually	Wrote the file loaders and buy(): checks seller, ownership and credit, moves the money, writes	Pass (runs 2 and 3)

Test	What it was testing	What was wrong with the output	Error in the code / test	What we changed	Result after fix
			read	the 04 record	
S04	Refund moves credit between accounts	Run 1: balances didn't change (buyer1 should be 510.00, seller1 490.00)	refund() and adjustCredit() were stubs	Wrote refund(): checks both accounts, moves the credit, writes the 05 record	Pass (runs 2 and 3)
S05	Deleted user's account and games can't be used	Run 1: deleted account still showed up in list	deleteUser() and removeUser() were empty stubs	Wrote deleteUser(): removes the account and the user's games for sale, writes the 02 record	Pass (runs 2 and 3)
S06	New game can't be bought same session; duplicate name rejected	Run 1: wrong records and error count	sell() was still a stub that always returned false	Wrote sell(): checks permission, price and name, writes the 03 record	Pass (runs 2 and 3)
S07	$1000 limit is per account	Run 1: credit limits weren't tracked per account (buyer1 and seller1 should both be 1500.00)	addCredit() and adjustCredit() were stubs, and nothing tracked the $1000-per-session limit	Wrote addCredit() with a per-account session total, writes the 06 record	Pass (runs 2 and 3)
S08	Creating each account type; duplicate name	Run 1: new accounts weren't created	createUser() was still a stub that always returned false	Wrote createUser(): checks the name and type, rejects duplicates, adds the account with 0 credit and writes the 01 record	Pass (runs 2 and 3)
S12	Games file missing	Run 1: login was accepted even though the games file was missing	loadGames() ignored the file path and returned an empty list	loadGames() now opens the file and login is rejected with a message if it can't be read	Pass (runs 2 and 3)
S13	Broken collections file	Run 1: login was accepted with a broken collections file	loadCollections() didn't check the record format	Added format checks for collection records and the END line	Pass (runs 2 and 3)
S14	Games file with no END line	Run 1: login was accepted with no END line in the games file	loadGames() didn't check for END	Missing END is now reported and the login is rejected	Pass (runs 2 and 3)
S15	Files using the handout's line widths	Run 1: the buy was rejected, so its 04 record was missing	buy() was a stub, and the games/collections files were never actually read	Loaders accept both widths; buy() implemented	Pass (runs 2 and 3)
S17	Credit limit resets at next login	Run 1: addcredit was rejected, so the 06 record was missing	addCredit() and adjustCredit() were stubs, and nothing tracked the $1000-per-session limit	Session totals reset and files reload at each login	Pass (runs 2 and 3)
S18	Buying with credit added in the same session	Run 1: couldn't buy with credit added earlier in the same session	addCredit() and buy() were stubs	Credit changes are applied in memory so later transactions see them	Pass (runs 2 and 3)
S21	Names with spaces; max-length names	Run 1: names with spaces / maximum-length names were rejected	createUser() and sell() were stubs	Names with spaces and exact max lengths now work	Pass (runs 2 and 3)
TC47	Search by Admin - Games and Users	Run 5: "search" was an unrecognized command, so no results were printed	SEARCH wasn't implemented yet (we wrote these tests first)	Added the search command: FrontEnd::handleSearch(), TransactionProcessor::search() and GameManager::isSelling()	Pass (run 6)
TC48	Search by Standard User Hides Non-Selling Admins	Run 5: "search" was an unrecognized command, so no results were printed	SEARCH wasn't implemented yet (we wrote these tests first)	Added the search command: FrontEnd::handleSearch(), TransactionProcessor::search() and GameManager::isSelling()	Pass (run 6)
TC49	Search Is Case-Insensitive for a Sell-Standard User	Run 5: "search" was an unrecognized command, so no results were printed	SEARCH wasn't implemented yet (we wrote these tests first)	Added the search command: FrontEnd::handleSearch(), TransactionProcessor::search() and GameManager::isSelling()	Pass (run 6)
TC50	Search With No Matches	Run 5: "search" was an unrecognized command, so no results were printed	SEARCH wasn't implemented yet (we wrote these tests first)	Added the search command: FrontEnd::handleSearch(), TransactionProcessor::search() and GameManager::isSelling()	Pass (run 6)
TC51	Search Before Login	Run 5: search before login gave "unrecognized command" instead of "you must log in"	SEARCH wasn't implemented yet (we wrote these tests first)	Added the search command: FrontEnd::handleSearch(),	Pass (run 6)

Test	What it was testing	What was wrong with the output	Error in the code / test	What we changed	Result after fix
				TransactionProcessor::search() and GameManager::isSelling()	
TC52	Blank Search Query	Run 5: “search” was an unrecognized command, so the blank query was never checked	SEARCH wasn’t implemented yet (we wrote these tests first)	Added the search command: FrontEnd::handleSearch(), TransactionProcessor::search() and GameManager::isSelling()	Pass (run 6)
TC53	Search After Deleting a User	Run 5: “search” was an unrecognized command, so no results were printed	SEARCH wasn’t implemented yet (we wrote these tests first)	Added the search command: FrontEnd::handleSearch(), TransactionProcessor::search() and GameManager::isSelling()	Pass (run 6)
TC54	Newly Listed Game Not Found by Search	Run 5: “search” was an unrecognized command, so no results were printed	SEARCH wasn’t implemented yet (we wrote these tests first)	Added the search command: FrontEnd::handleSearch(), TransactionProcessor::search() and GameManager::isSelling()	Pass (run 6)
H01	Test script on Windows (Git Bash)	run_tests.sh failed with “Permission denied” on python3	Our script found the Windows Store python3 shortcut, which does not run	Script checks that python3 actually runs before using it, otherwise uses python	Fixed (run 3). The scripts no longer use Python.
TC01	Valid Login	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn’t show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC02	Invalid Login	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn’t show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC03	Transaction Before Login	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn’t show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC04	Second Login During Session	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn’t show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC05	Valid Logout	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn’t show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC06	Logout Before Login	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn’t show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC07	Transaction After Logout	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn’t show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC08	Valid Create User	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn’t show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC09	Unauthorized Create User	Run 7: last line of the .atf was just 00, expected `00 buyer1 BS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn’t show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC10	Duplicate Username	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn’t show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC11	Username Over 15	Run 7: last line of the .atf was just 00, expected `00 admin`	logout() wrote a bare 00, and the program	logout() now writes the full 00 record (user,	Pass (run 8)

Test	What it was testing	What was wrong with the output	Error in the code / test	What we changed	Result after fix
	Characters	`AA 005000.00`. The .out was also missing the typed input	doesn't show the input it reads from a file	type, credit). Input lines are echoed when they come from a file	
TC12	Username Exactly 15 Characters	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC13	Valid Delete User	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC14	Unauthorized Delete User	Run 7: last line of the .atf was just 00, expected `00 buyer1 BS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC15	Delete Nonexistent User	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC16	Delete Current User	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC17	Valid Sell	Run 7: last line of the .atf was just 00, expected `00 seller1 SS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC18	Buy-Standard User Attempts Sell	Run 7: last line of the .atf was just 00, expected `00 buyer1 BS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC19	Maximum Game Price	Run 7: last line of the .atf was just 00, expected `00 seller1 SS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC20	Price Above Maximum	Run 7: last line of the .atf was just 00, expected `00 seller1 SS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC21	Game Name Exactly 25 Characters	Run 7: last line of the .atf was just 00, expected `00 seller1 SS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC22	Game Name Over 25 Characters	Run 7: last line of the .atf was just 00, expected `00 seller1 SS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC23	Duplicate Game Name	Run 7: last line of the .atf was just 00, expected `00 seller1 SS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC24	Invalid Sell Input	Run 7: last line of the .atf was just 00, expected `00 seller1 SS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC25	Transaction on Newly Listed Game	Run 7: last line of the .atf was just 00, expected `00 seller1 SS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC26	Valid Buy	Run 7: last line of the .atf was just 00, expected `00 buyer1 BS 000485.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC27	Sell-Standard User Attempts Buy	Run 7: last line of the .atf was just 00, expected `00 seller1 SS 000500.00`. The .out was also missing the	logout() wrote a bare 00, and the program	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they	Pass (run 8)

Test	What it was testing	What was wrong with the output	Error in the code / test	What we changed	Result after fix
		typed input	doesn't show the input it reads from a file	come from a file	
TC28	Buy Nonexistent Game	Run 7: last line of the .atf was just 00, expected `00 buyer1 BS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC29	Insufficient Credit	Run 7: last line of the .atf was just 00, expected `00 lowcredit BS 000005.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC30	Exact Credit Purchase	Run 7: last line of the .atf was just 00, expected `00 exactbuyer BS 000000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC31	Already-Owned Game	Run 7: last line of the .atf was just 00, expected `00 buyer1 BS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC32	Valid Refund	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC33	Unauthorized Refund	Run 7: last line of the .atf was just 00, expected `00 buyer1 BS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC34	Refund to Nonexistent Buyer	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC35	Refund from Nonexistent Seller	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC36	Valid Add Credit - Standard User	Run 7: last line of the .atf was just 00, expected `00 buyer1 BS 000600.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC37	Valid Add Credit - Admin	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC38	Admin Adds Credit to Nonexistent User	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC39	Add Exactly $1000	Run 7: last line of the .atf was just 00, expected `00 buyer1 BS 001500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC40	Add More Than $1000	Run 7: last line of the .atf was just 00, expected `00 buyer1 BS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC41	Multiple Credit Additions to $1000	Run 7: last line of the .atf was just 00, expected `00 buyer1 BS 001500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC42	Multiple Credit Additions Over $1000	Run 7: last line of the .atf was just 00, expected `00 buyer1 BS 001100.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC43	Invalid Credit Amount	Run 7: last line of the .atf was just 00, expected `00 buyer1 BS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they	Pass (run 8)

Test	What it was testing	What was wrong with the output	Error in the code / test	What we changed	Result after fix
				come from a file	
TC44	Invalid Transaction Code	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC45	Daily Transaction File End Code	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC46	Daily Transaction File Formatting	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC47	Search by Admin - Games and Users	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC48	Search by Standard User Hides Non-Selling Admins	Run 7: last line of the .atf was just 00, expected `00 buyer1 BS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC49	Search Is Case-Insensitive for a Sell-Standard User	Run 7: last line of the .atf was just 00, expected `00 seller1 SS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC50	Search With No Matches	Run 7: last line of the .atf was just 00, expected `00 fullgamer FS 000300.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC51	Search Before Login	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC52	Blank Search Query	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC53	Search After Deleting a User	Run 7: last line of the .atf was just 00, expected `00 admin AA 005000.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
TC54	Newly Listed Game Not Found by Search	Run 7: last line of the .atf was just 00, expected `00 seller1 SS 000500.00`. The .out was also missing the typed input	logout() wrote a bare 00, and the program doesn't show the input it reads from a file	logout() now writes the full 00 record (user, type, credit). Input lines are echoed when they come from a file	Pass (run 8)
S01-S11, S15, S17-S19, S21	Extra tests that write a .atf	Run 7: 16 of 21 failed on the last line (bare 00)	Same 00 problem as above	Same fix; added the expected 00 line to each case	Pass (run 8)